

Vapnik - Chervonenkis.

* VC dimension

→ used in ML to measure complexity of a model's hypothesis space. It gives us an idea of how flexible or expressive the model is.

For a hypothesis class, VC dimension is defined as largest no. of points that class can shatter, where 'shatter' means class can realize all possible labelings of those points.

Consider set of pts in 2D space. We want to determine VC dimension of set of all possible lines that can classify these points.

The line can separate pts into 2 regions those above line & those below line.

To simplify let's consider 3 points in general position in 2D space.

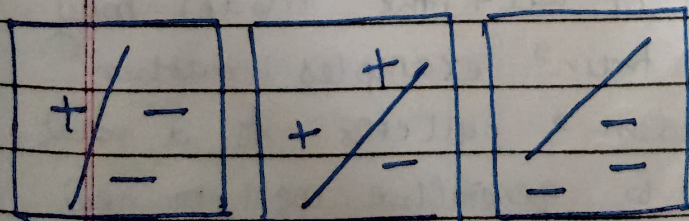
There are 8 labelings of these points (2³), represents possible values of whether each pt is below or above line.

Let's analyze if a line can shatter these three points.

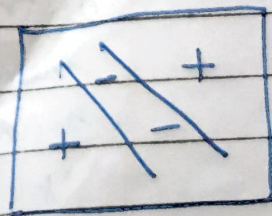
- 1) If all 3 pts are on same side of line, the line can correctly classify them
- 2) If two pts (2) are on one side of line & one point is on other side, the line can correctly classify them
- 3) If each pt is on diff side of line, the line can correctly classify them

So, the set of all possible lines can shatter any set of 3 pts in general position, but it cannot shatter any set of 4 pts.

Therefore, VC d of line in 2D space is 3



3 points shattered



4 pts impossible

Catalyst of Overfitting.

⇒ The catalyst are factors that ~~attribute~~ contribute to ML model learning training data too well, to the extent that it performs poorly on unseen data.
Here are some catalyst:-

1) Complexity of Model:-

When the model is complex relative to the size & complexity of training data, it can capture noise or random fluctuations in data. This often leads to poor generalization to new, unseen data.

2) Insufficient Training Data:-

If training data is too small or does not adequately represent underlying distribution of data, the model may memorize training examples rather than underlying patterns. As a result, it fails to generalize well to new data.

3) Data Leakage:-

occurs when inf. from test set or future data is used inadvertently during model training. This can artificially inflate model's performance on training data & lead to overfitting.

4) Improper Cross-Validation:-

Inadequate or improper cross-validation technique can also contribute to overfitting.

5) Feature Overfitting:-

Including irrelevant or too many features in model can lead to overfitting.

* Hypothesis testing in real-world applicⁿ

→ Statistical method used to make inferences about a population parameter based on sample data.

It involves formulating a null hypothesis (H_0) & alternative hypothesis (H_1), collecting data, and using statistical technique to determine whether there is enough evidence to reject the null hyp in favour of alternative hyp.

Applicatⁿ:-

1) Medicine & Healthcare:-

In clinical trials, researchers use hyp test^s to determine whether a new treatment is more effective than existing one. They collect data on patient outcome & conduct hypothesis tests to assess whether there is significant diff. in effectiveness betⁿ treatments.

2) Finance & Economics:-

used in finance to test theories about stock market behaviour, asset pricing models, & impact of economic policies.

3) Environmental Science:-

Researchers use hyp tests to assess the effects of environmental factors on ecosystems

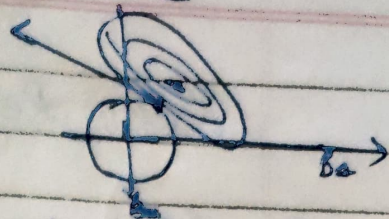
4) Education & Psychology:-

used in educational research to evaluate effectiveness of teaching methods or interventions.

ridge estimate

 B_2

Ridge Regression:-

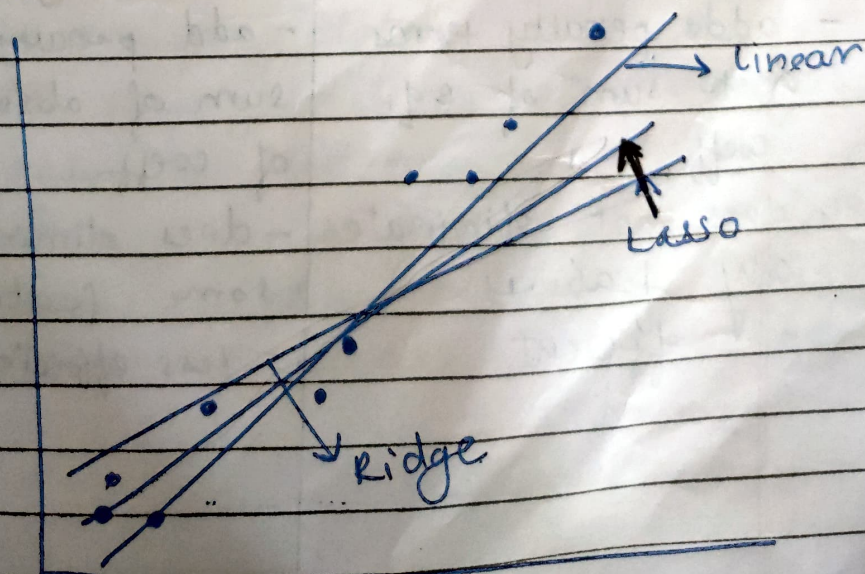


Sometimes, we ~~must~~ try to fit a line to our data points (like predicting house prices based on size and location), we end up with a line which is too wiggly.

This is because, the model is trying too hard to fit each data point perfectly, even if it's just noise.

So, Ridge Reg helps by adding a small extra term to our equation that penalizes big coefficients.

So, if model wants to make a coeff. really big just to fit one outlier, it'll ~~to~~ think twice because it'll have to pay a penalty for it.



Lasso Reg:



Similar to Ridge Reg, but sometimes we want even simpler soln.

We not only want to prevent coeff from getting too big, but we also want some of them to be exactly zero.

This helps in dropping less imp. features from our model. So, Lasso Reg does exactly that. It not only penalizes big coeffs, but can also force some of them to be zero.

This makes model, simpler & easier.

-

Ridge	Lasso
- shrinks coeff toward zero	- encourages coeff to be exactly zero
- adds penalty terms to sum of sq. coeff	- add penalty to & to sum of absolute values of coeff
- does not eliminate any features	- does eliminate some features.
- efficient	- less efficient

KNN
or

classmate

Date _____

Page _____

Nearest Neighbor Classification

- simple but effective ML model for classification tasks.
- It works on the principle of finding nearest data points to a given sample & classify^g them on majority class.

Work :-

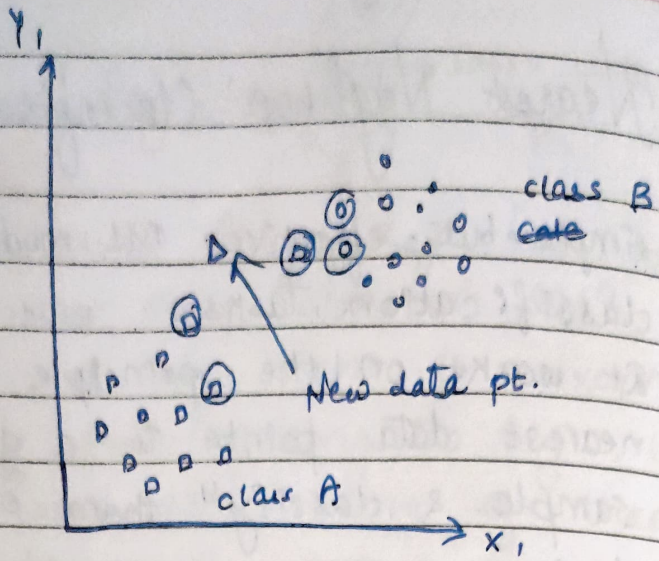
⇒ Training Phase:-

alg simply memorizes all data pts & their correspond^g class labels.

ol classificatⁿ phase:-

- When a new sample is presented for classificatⁿ, alg calc distance betⁿ sample and all other samples in training data.
- It then selects k-nearest neighbour based on distances
- Finally, it assigns class label to new sample based on majority class of its k nearest neighbors.

Before KNN

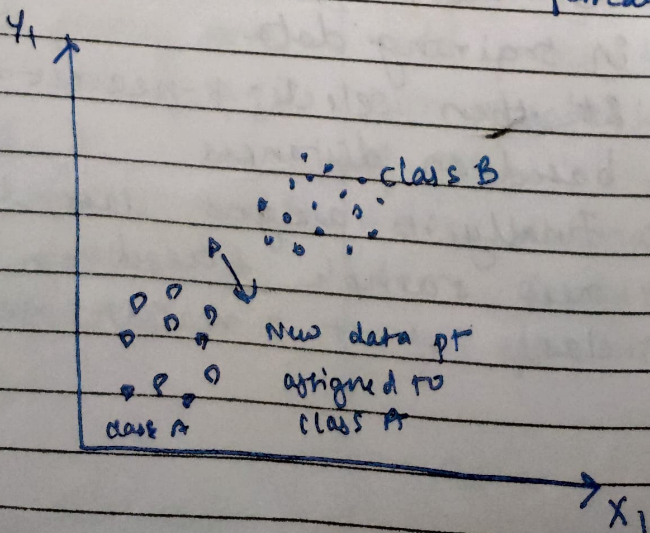


Eg:-

Suppose we have a dataset of people's heights & weights, along with their genders. You want to classify a new person as Male or Female based on height & weight.

- You first find nearest person in dataset to new person.
- If nearest p is male, you classify it as male or it is female.

After KNN



* k-mean

- takes mean of data points to create new pts called centroids

- Centroids are new pts previously not found in data.

- used for numerical data

- uses Euclidean dist.

- faster

- less costly

k-medoids cluster^g.

- uses points from data to serve as pts called as medoids.

- Medoids are existing points from the data.

- used for both Num & categorical

- Manhattan dist.
 $|x_1 - x_2| + |y_1 - y_2|$

- comparatively slower.

- more costly.

Amli
Amli

* Hierarchical clustering

- used in unsupervised ml.
- particularly useful when you want to know the inherent structure of your data, especially when you don't have predefined labels.
- builds a hierarchy of clusters by either recursively merging smaller clusters into large ones (agglomerative) or large into smaller ones (divisive).

Work :-

1) Agglomerative Approach:-

- Starts by considering each data pts as a single cluster
- Merge two clusters into larger cluster
- Repeat process, merging closest clusters, until there is only one cluster containing all data points.

2/ Divisive Approach:

- Start with cluster containing all data points.
- split cluster into smaller clusters.
- continue splitting until each cluster contains only one data point.

Use cases:-

- used in biology for classifying species based on genetic similarity
- 'market segmentatⁿ' to group customers based on similarities in purchas^d behaviour

Minority class

- class with fewer instances or observations compared to others
- often represents class of interest or target class that we want to predict or analyze
- crucial for imp tasks like fraud detection or medical diagnosis

Methods like resampling or algorithmic adjustments used to address imbalances.

Impurity

- contains mixture of diff. classes.
- indicates lack of clarity, making it difficult to classify accurately
- essential for improving overall accuracy by reducing noise & ambiguity.

- Techniques like ensemble methods or boosting algorithms helps in reducing impurity.

Distance-based

- models classify data pts based on their proximity using distance metrics.

- eg Alg - KNN

- intuitive & conceptually simple.

- Sensitive to irrelevant features

- eg - Predicting a person's income level based on their age & education with KNN.

Tree based models.

- models that make decision based on feature values

- Decision Trees

- Easy to interpret & visualize

- prone to overfitting especially with deep trees.

- Predicting customer churn based on demographic & behavioural data using Decision trees

Gini Index:-

- proportion of impurity or inequality
- In ml, it is utilized as an impurity measure in decision-tree algorithm for classificatⁿ task.
- it is utilized to determine best feature to split the data on at every node of the tree.

$$\text{Gini} = 1 - \sum_{i=1}^n (p_i)^2$$

p_i = probability of thing with a class.

for eg:-

Consider a binary classificatⁿ issue, where we have a dataset of 10 examples with two classes "Positive" and "Negative".

Out of 10 examples, 6 have a place with "Positive" class and 4 have a place with "Neg" class.

To calc Gini Index, we calc prob.

$$p_1 = 6/10 = 0.6, \quad p_2 = \frac{4}{10} = 0.4.$$

$$\begin{aligned} \text{Gini}(S) &= 1 - (p-1) \cdot (p-2) \\ &= 1 - [(0.6)^2 + (0.4)^2] \\ &= 0.48 \end{aligned}$$

Adv:-

- computationally efficient
- Good for Binary classificatⁿ
- less prone to overfitting.

Disadv:-

- Not good for continuous var
- Ignores features interactⁿ
- Not ideal for some dataset.

ways to prevent overfitting.

1) Cross-validation
assess model's performance.

2) Regularization
 L_1 (Lasso) L_2 (Ridge) regularization
add a penalty terms to model,
discourage overly complex models
with large coeff.

3) Feature selection:-
- select relevant features
- reduce model's complexity.

4) Ensemble methods
Em like bagging, boosting &
random forests helps improve
performance.

5) Simpler Model:-
Simple models with fewer
parameters.

Multivariate Linear Regression

- one dependent & multiple inde var.
- Problem statement
Suppose we want to predict student's final exam score based on factors such as study hours, attendance, previous exam scores.
- Gather Data:-
For each student record no. of study hr (x_1), attendance $\%$ (x_2), prev exam (x_3), (x_4), etc) as well final exam score ~~(x_5)~~ (y).
- Data preparatⁿ
Org data into a table...
- Model formulatⁿ:-

$$y = B_0 + B_1x_1 + B_2x_2 + B_3x_3 + \dots + E.$$

↓
dependent

B_0 = intercept
 x_1, x_2, x_3 → inde var
- Model training
- Model Evaluatⁿ
- Interpretatⁿ

Entropy :-

- ⇒ It is measure of impurity or uncertainty in a dataset.
- ⇒ It is used to quantify impurity of a dataset before & after splitting the data.
- ⇒ Mathematically, Binary classification problem (Two classes such as 'Yes' & 'No') the entropy is calc as,
$$\text{Entropy}(S) = -P_{\text{pos}} \log_2(P_{\text{pos}}) - P_{\text{neg}} \log_2(P_{\text{neg}})$$
 - $P_{\text{pos}} \rightarrow$ proportion of +ve instances (Yes)
 - $P_{\text{neg}} \rightarrow$ " of -ve " (No)
- ⇒ Entropy ranges from 0 to 1.
 - low entropy - pure dataset
 - high entropy - impurity.

Use Cases:-

1) Thermodynamics:-

- entropy is measure of disorder or randomness in a system.
- used to understand behaviour of system in thermal equilibrium.

2) Machine Learning.

- measure impurity in decision
- tree particularly in classification

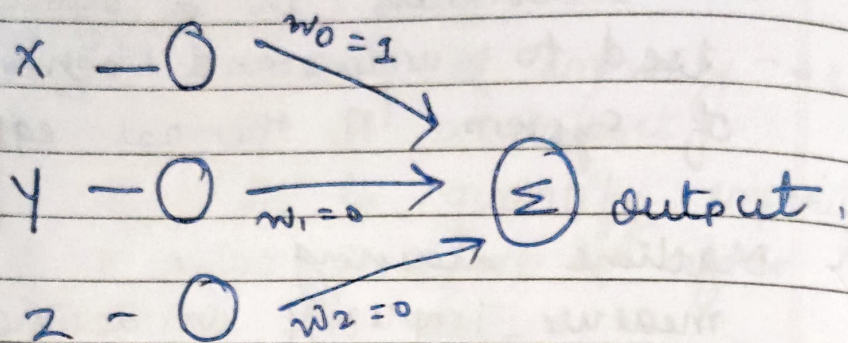
3) Quantum Mechanics:-

- understand^g behaviour of quantum systems

4) Informatⁿ Theory:-

- avg amt of inf produced by a random variable.

Perceptron :- ISA



- fundamental concept in artificial neural network developed by Frank Rosenbelt in 1957.
- It is a binary linear classifier, it takes multiple binary inputs & single produce single binary output.

Here's how it works:-

1. Inputs:- (I)

Each input is associated with a weight that signifies its imp. This inputs are multiplied by signified respective weights.

2. Summation:- The weighted inputs

are summed together with an added bias term.

3) Activation funcⁿ:-

Sum is then passed through an activation funcⁿ, typically a step function.

If result is above a certain threshold, perceptron outputs 1; otherwise, outputs 0.

Applicatⁿ:-

1) Binary classificatⁿ
where inputs can be classified into two categories for spam-email detection, medical diagnosis.

2) NLP:-

used for tasks such as sentiment analysis, text classificatⁿ, part of-speech tagging.

3) Control systems:-

for tasks like robotics or automated systems.

4) Financial forecast^g:-
tasks like stock market
prediction, credit scoring, or
fraud detection.

$$TP = 189$$

$$FP = 99$$

$$TN = 901$$

$$FN = 38$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{189}{189 + 99} = \frac{189}{288} = 0.656$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$= \frac{189}{189 + 38} = \frac{189}{227} = 0.832$$

Accuracy

$$F1 \text{ Score} = \frac{2 \times 0.656 \times 0.832}{0.656 + 0.832}$$

$$= \frac{1.117824}{1.488}$$

$$= \underline{\underline{0.75122}}$$